

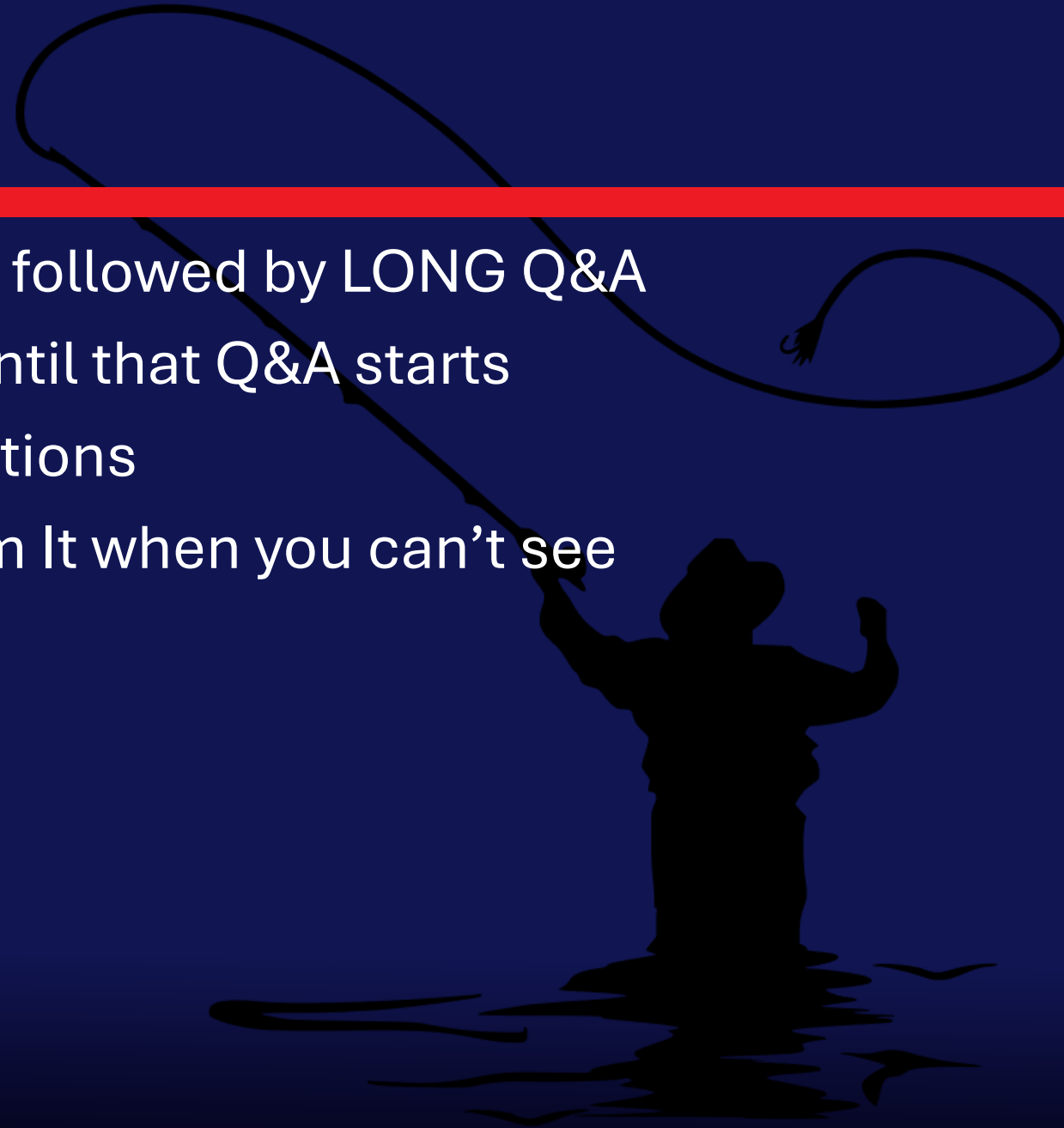
Build Your Test Lab:

No Lab? No Problem!



Attention

- ◆ MMS sessions are 60-75 minutes followed by LONG Q&A
- ◆ Please hold detailed questions until that Q&A starts
- ◆ Feel free to ask clarification questions
- ◆ Remind the speakers to use Zoom It when you can't see



Speakers



Mattias Benninge

Principal Engineer
2Pint Software

 @matbg



Mike Terrill

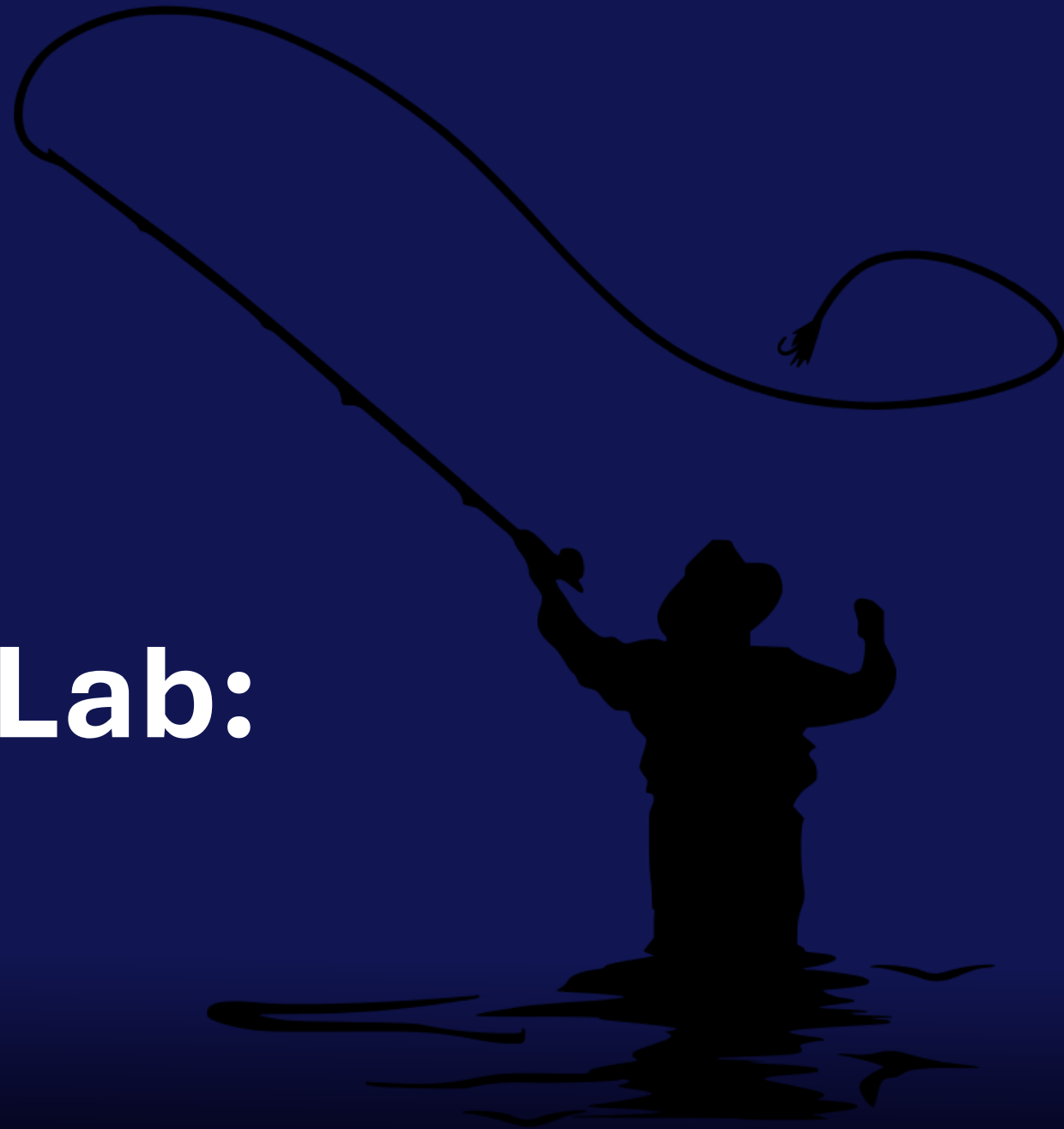
Principal Engineer, Microsoft MVP
2Pint Software

 @miketerrill

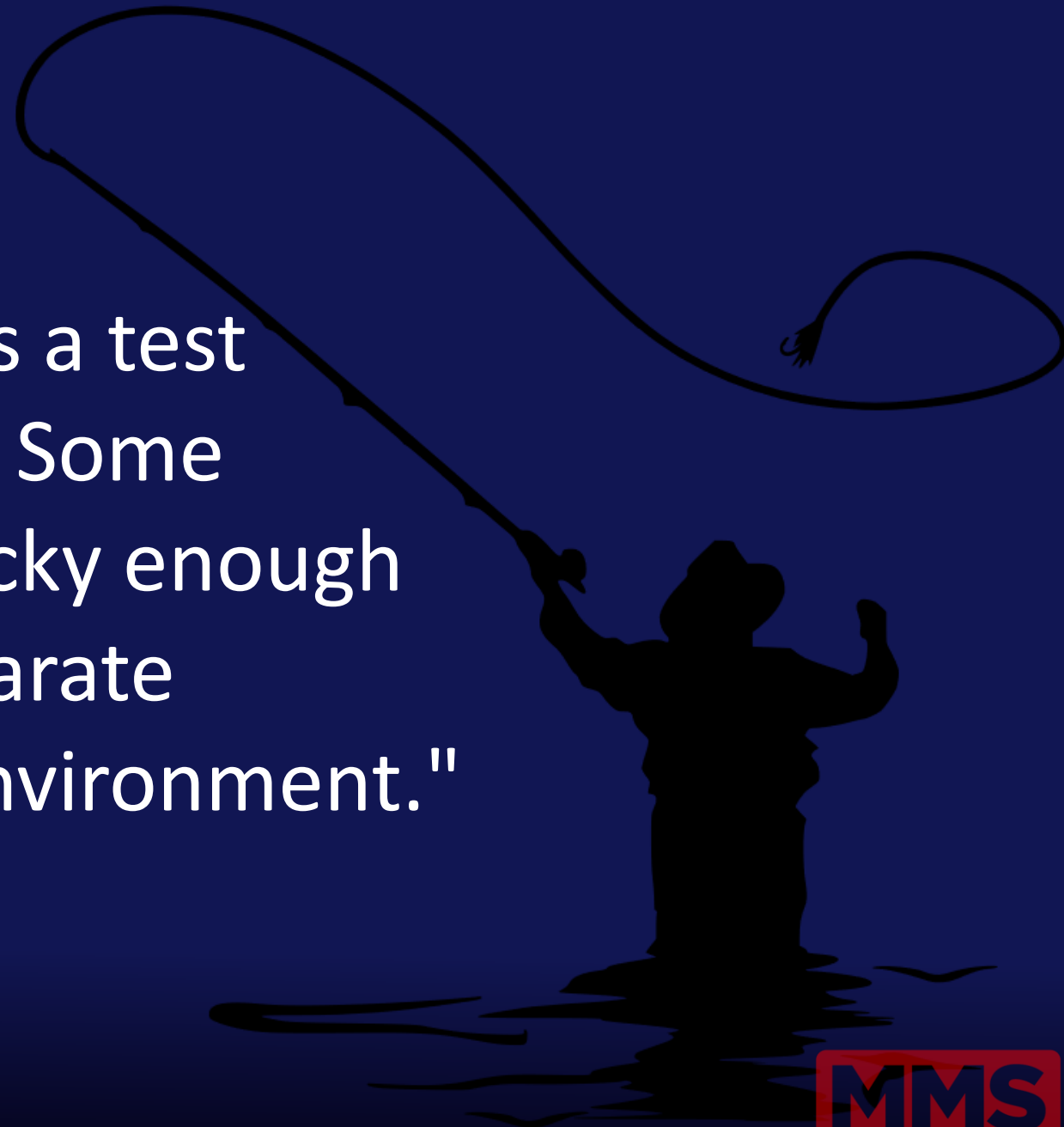
Can't see our slides? Can't hear? Need to repeat the question? Call us out!

Build Your Test Lab:

No Lab? No Problem!



"Everyone has a test environment. Some people are lucky enough to have a separate production environment."



Why You Should Have a Lab

- ◆ Never, ever test in production! (RGE)
- ◆ Work Labs
 - ◆ Should replicate production
 - ◆ Good hardware collection
 - ◆ Test here first
- ◆ Home Labs
 - ◆ Hone your skills, learn new technology
 - ◆ Virtual machines are perfect



Lab Software

- ◆ Hydration Kit by Johan
 - ◆ <https://www.deploytoresearch.com/hydration-kit-for-windows-server-2022-sql-server-2019-and-configmgr-current-branch/>
 - ◆ Advantages – Customize to your environment.
 - ◆ Disadvantages – Takes some time to setup.
- ◆ Windows and Office 365 Deployment Lab Kits from Microsoft
 - ◆ Windows 11: <https://www.microsoft.com/en-us/evalcenter/evaluate-windows-11-office-365-lab-kit>
 - ◆ Advantages – Setup quickly.
 - ◆ Disadvantages – Hardcoded to corp.Contoso.com
- ◆ Build Your Own



Low end (OSD) vs High End (OSD)

Single Virtual Machine

- ◆ Use your existing Infrastructure
- ◆ Create a Pre-Prod Task Sequence
- ◆ Test the Client on a Virtual Machine
- ◆ Virtual Machines are great for testing!

Full Lab Environment

- ◆ Full Virtualized Environment
 - ◆ Domain Controller
 - ◆ Gateway/Router
 - ◆ ConfigMgr Server
 - ◆ Distribution Point
 - ◆ Test Clients
- ◆ Augmented w/ Physical Clients

Hardware

◆ Processors:

- ◆ i7 or Newish i5
- ◆ i7 8th or better for Windows 11
- ◆ Used Xeon Scalable are good buy
- ◆ Ryzen/ThreadRipper
- ◆ 8 Cores – 16 threaded
- ◆ See: <https://cpubenchmark.net>

◆ RAM:

- ◆ 32GB (min) 64GB+ (Optimal)
- ◆ Servers: 64GB+

◆ Storage:

- ◆ 1tb NVME SSD - King for VM's
- ◆ Put your code data on HDD
- ◆ See: <https://shucks.top>
- ◆ See: <https://diskprices.com>

◆ Networking:

- ◆ 1Gbps (min)
- ◆ 10GB SFP+ Used ConnectX-3 w/ DAC - Good 4 testing
- ◆ 2.5gb is the new thing
<https://amazon.com/s?k=2.5gb+switch>

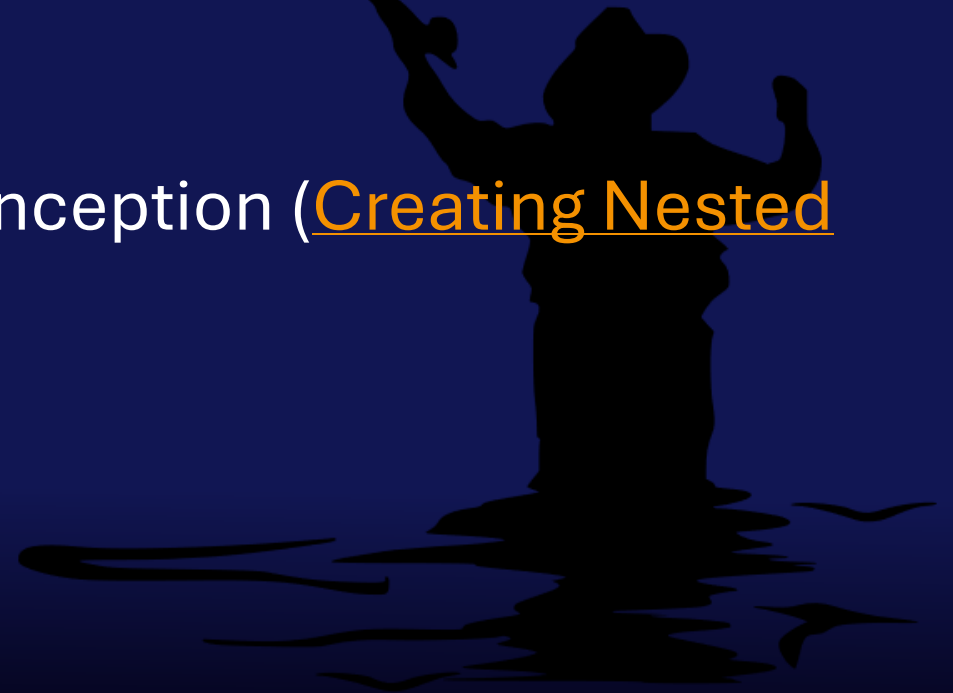
Host Examples

- ◆ Cheap – Upgrade
 - ◆ Re-Use an old Gaming PC, add an extra NVME drive.
 - ◆ Add some more RAM Existing Laptop or Desktop!
- ◆ Moderate - Used
 - ◆ Search on E-Bay – The [GaryTown](#) Method
 - ◆ Search for Workstations on E-Bay or <https://PCServerAndParts.com>
 - ◆ Get used PC's with CPU and RAM. Get your own disks.
- ◆ All New
 - ◆ Super Small - [Minisforum MS-01](#)
 - ◆ [Fanless](#) Firewall Appliances w/ Intel n100



Setting up a Virtual Machine

- ◆ 4 CPU's ([Installing updates during Autopilot: Windows 11 edition, revisited again – Out of Office Hours \(oofhours.com\)](#))
- ◆ Dynamic vs Static Memory
- ◆ Checkpoints
- ◆ Secure Boot
- ◆ Nested Virtual Machines – aka Hyper-V inception ([Creating Nested Hyper-V VMs - Deployment Research](#))



Software

◆ Network Software

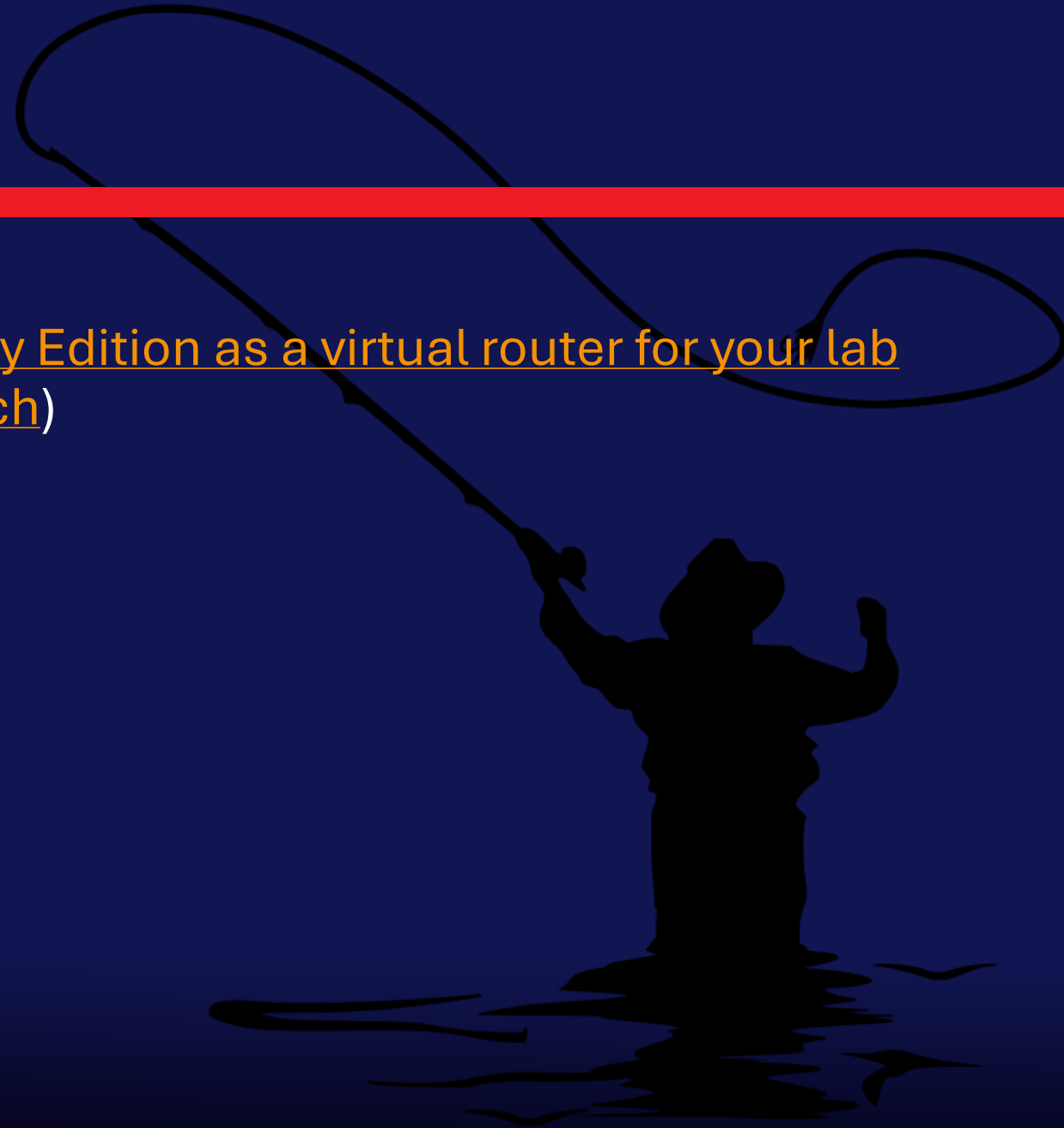
- ◆ pfSense ([Using pfSense Community Edition as a virtual router for your lab environment - Deployment Research](#))
- ◆ TailScale
- ◆ TinyPilot (KVM)

◆ HyperVisors

- ◆ Hyper-V
- ◆ VMWare
- ◆ ProxMox

◆ Storage

- ◆ Storage Spaces



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DEMO

Building your own lab



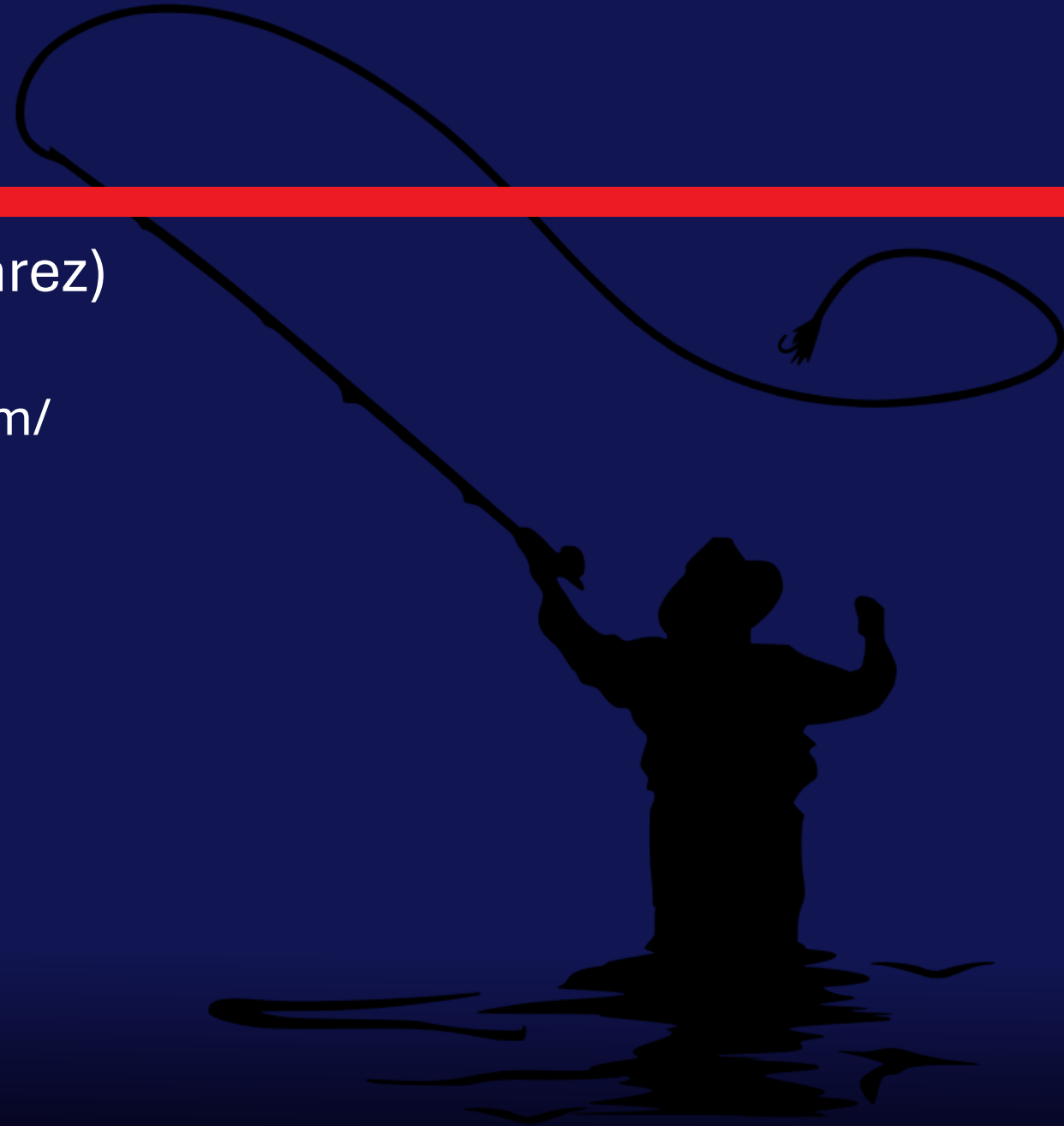
Intune LAB Tenant

- ◆ 30 day trial
 - + Free
 - Only last 30 days...
- ◆ Microsoft 365 Developer Program
 - + 25 Free licenses
 - Need a Visual Studio Pro (or better) Subscription
 - Not renewed unless doing active development
 - Pay as you go subscription not available (No MCC)
- ◆ Payed
 - + Keep as long as you pay
 - Cost



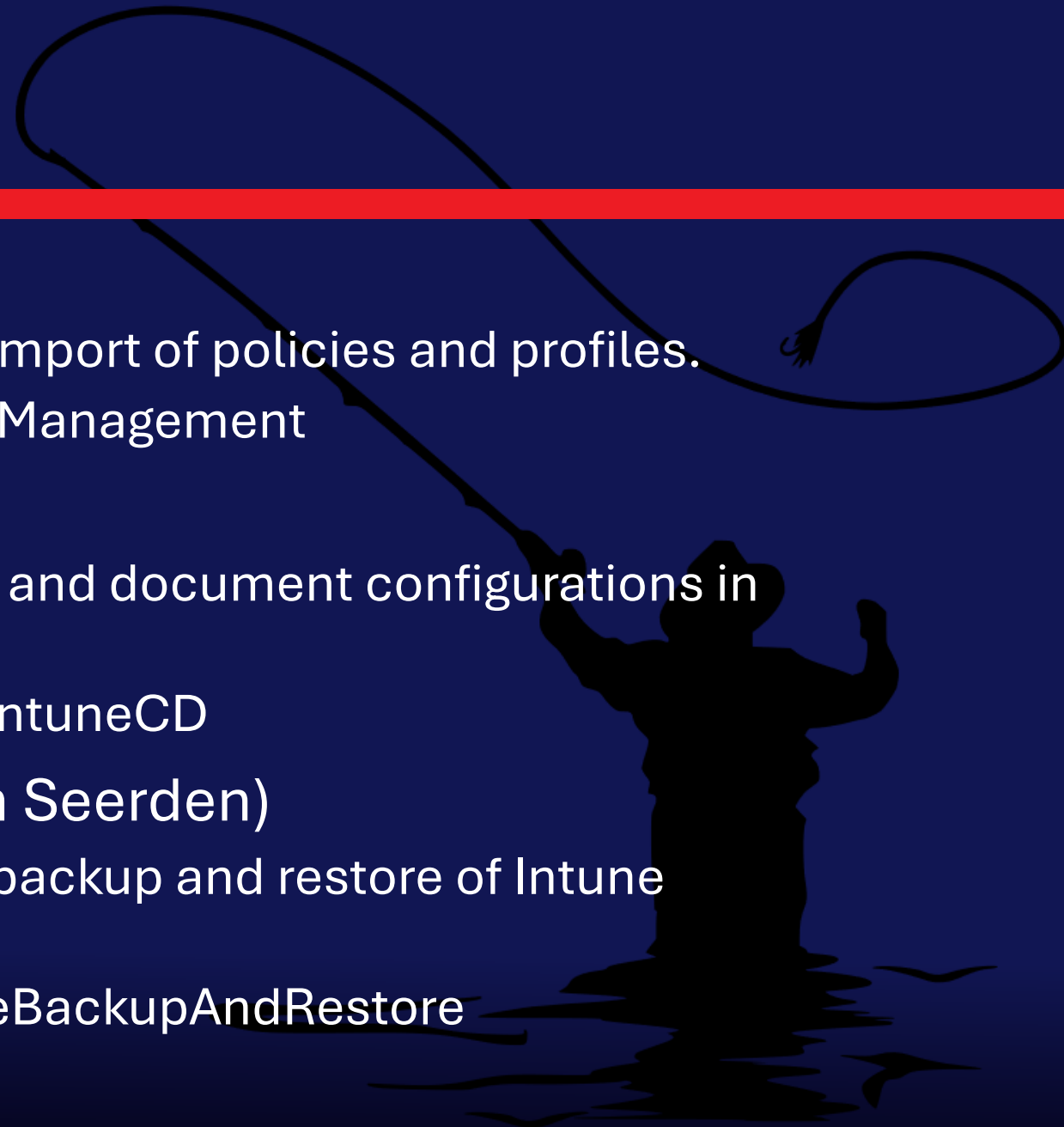
Intune Tools

- ♦ Intune Hydration Kit (by Jorge Suarez)
 - ♦ Good if starting with a fresh lab
 - ♦ <https://www.intunehydrationkit.com/>



Intune Tools

- ◆ Intune Management (by Micke K)
 - ◆ PowerShell tool to simplify Export/Import of policies and profiles.
 - ◆ <https://github.com/Micke-K/IntuneManagement>
- ◆ IntuneCI (by Tobias Almén)
 - ◆ Automation tool to backup, update and document configurations in Intune.
 - ◆ <https://github.com/almenscorner/IntuneCD>
- ◆ Intune Backup & Restore (by John Seerden)
 - ◆ Multitenant PowerShell module to backup and restore of Intune configurations.
 - ◆ <https://github.com/jseerden/IntuneBackupAndRestore>



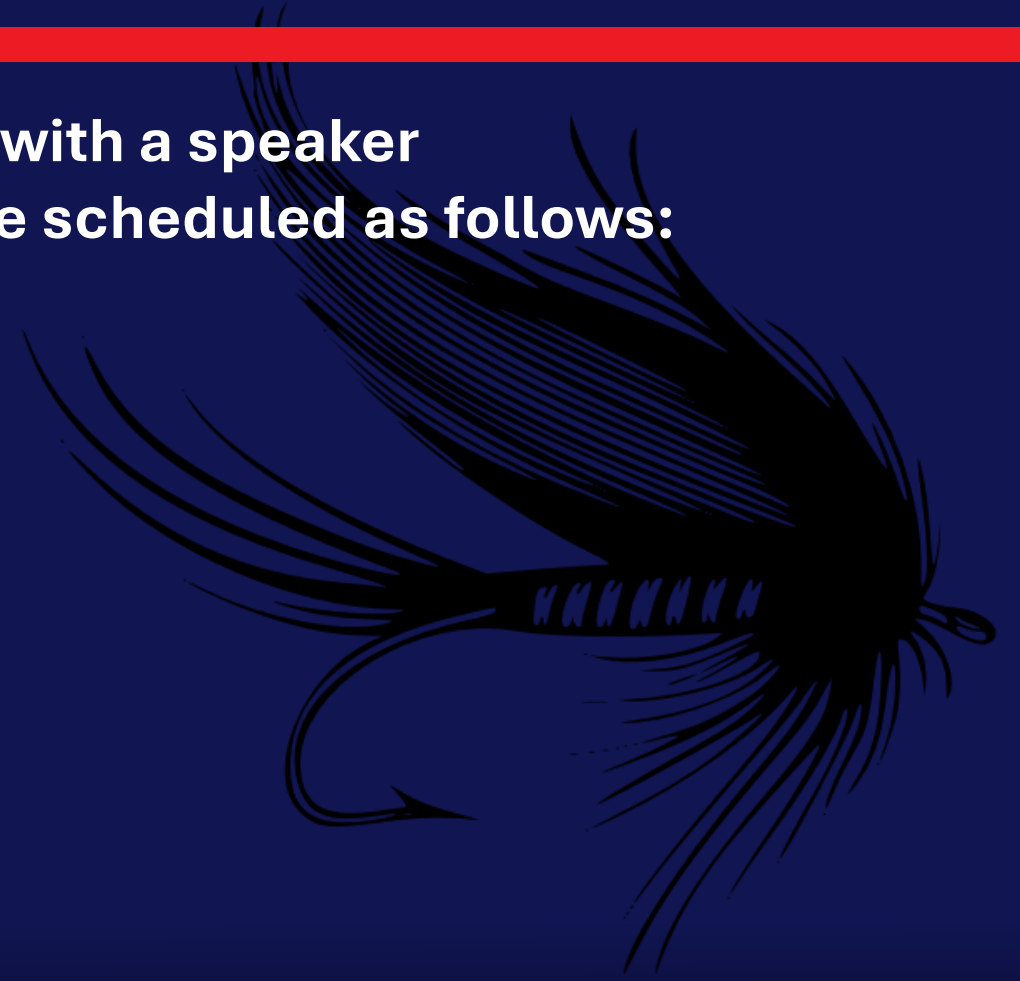
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Fishing Sessions

Be sure to find Wally and sign up for 1:1 time with a speaker
If you'd like to "catch" us for a session, we are scheduled as follows:



Don't forget to leave feedback for your speakers!

SAVE THE DATES

Oct 25-28, 2026



May 2-6, 2027



Oct 10-13, 2027



Extended Q&A



Recast

